

UBER DEMAND-SUPPLY GAP ANALYSIS

Executive Summary

The purpose of this project is to analyze Uber ride request data and identify the factors contributing to the demand-supply gap. The analysis focuses on ride completion rates, cancellations, driver availability, and customer demand patterns. Python was used for data cleaning and exploratory data analysis, SQL was used for business query analysis, and Excel was used to create an interactive dashboard. The findings revealed significant driver shortages during peak hours and airport pickup locations, resulting in a large number of unsuccessful ride requests.

1. Introduction

Uber is one of the world's leading ride-hailing platforms. Customer satisfaction depends heavily on driver availability and successful ride completion. However, customers often face issues such as ride cancellations and unavailability of drivers.

This project aims to analyze Uber ride request data to understand the reasons behind ride failures and provide actionable recommendations to improve operational efficiency and customer experience.

2. Business Problem

Uber faces the following challenges:

- Ride cancellations by drivers
- No Cars Available situations
- Driver shortages during peak demand periods
- Customer dissatisfaction due to unsuccessful ride requests

These challenges create a demand-supply gap where customer demand exceeds the available driver supply.

3. Project Objectives

The objectives of this project are:

- Analyze Uber ride request data
- Identify demand-supply gaps
- Understand ride cancellation patterns
- Evaluate driver availability

- Discover demand trends across time periods
 - Provide business recommendations to improve ride completion rates
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4. Dataset Description

The dataset contains 6745 ride requests and includes the following fields:

Column Name	Description
Request ID	Unique ride request identifier
Pickup Point	Location of pickup (Airport or City)
Driver ID	Driver identifier
Status	Ride status
Request Timestamp	Ride request date and time
Drop Timestamp	Ride completion date and time

Status Categories:

- Trip Completed
 - Cancelled
 - No Cars Available
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5. Data Cleaning and Preprocessing

The following preprocessing activities were performed using Python:

- Imported dataset using Pandas
- Converted timestamps into datetime format
- Identified and analyzed missing values
- Created Request Hour feature
- Created Time Slot feature
- Standardized data types
- Retained business-valid missing values

Missing Value Analysis

Driver ID contained missing values because no driver was assigned for requests marked as "No Cars Available".

Drop Timestamp contained missing values because rides were either cancelled or never completed.

These missing values were business-valid and retained for analysis.

6. Feature Engineering

Additional columns were created to support analysis.

Request Hour

Extracted from Request Timestamp to identify demand trends by hour.

Time Slot

Hours were grouped into business-friendly categories:

- Early Morning
- Morning
- Afternoon
- Evening
- Night

These features helped identify peak demand periods.

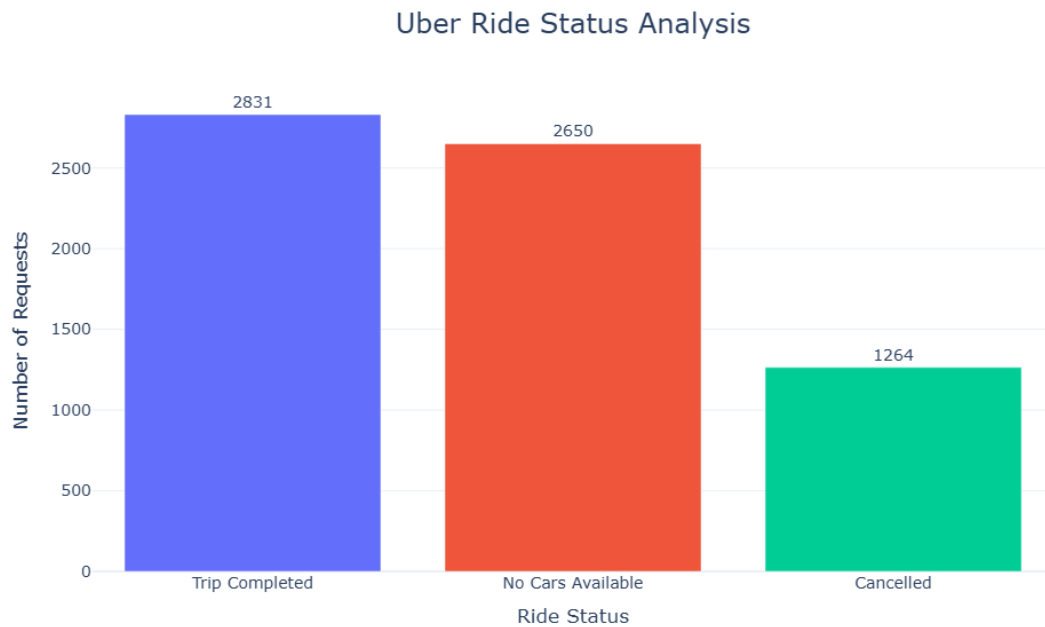
7. Exploratory Data Analysis (EDA)

* Ride Status Distribution

The analysis revealed three major ride outcomes:

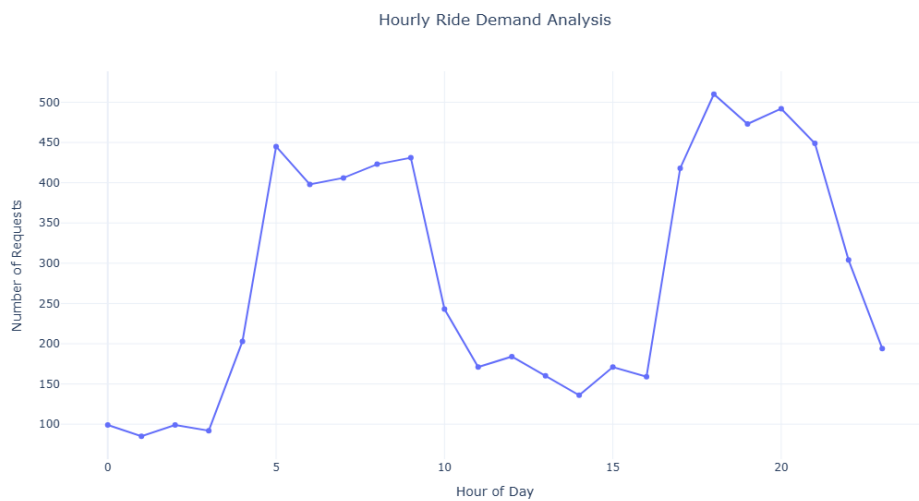
- Trip Completed
- Cancelled
- No Cars Available

A significant portion of ride requests were unsuccessful, indicating operational challenges.



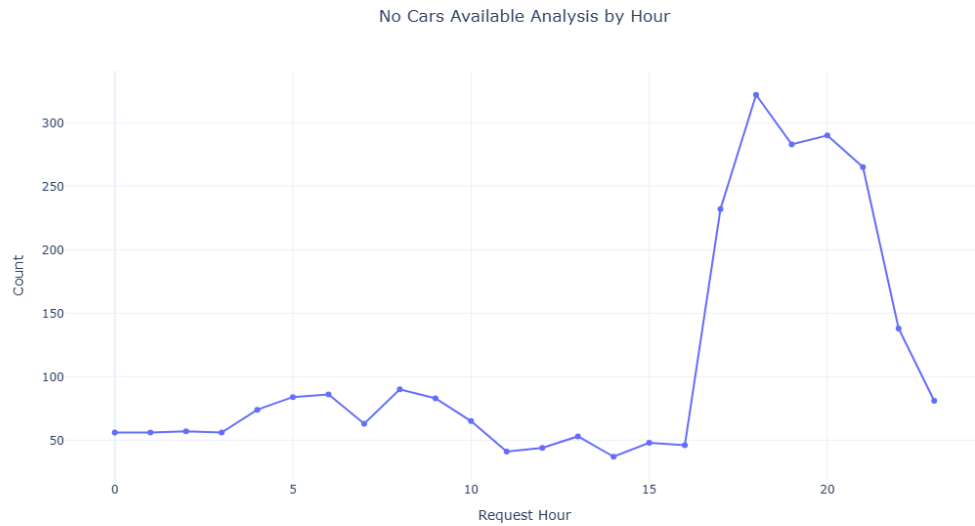
* Hourly Ride Demand Analysis

Ride requests were analyzed by hour to identify peak demand periods.



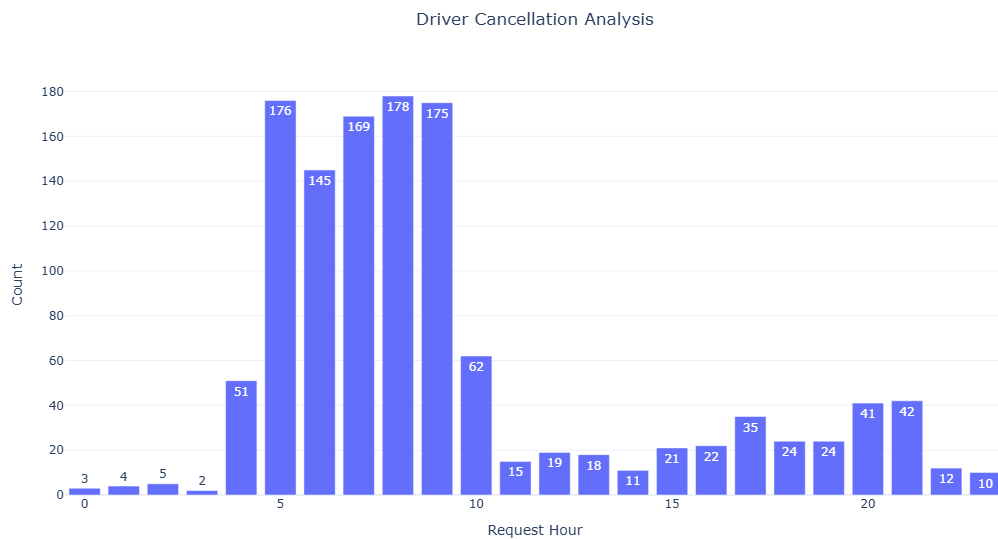
* No Cars Available Analysis

Analysis of unsuccessful requests highlighted periods where driver supply was insufficient.



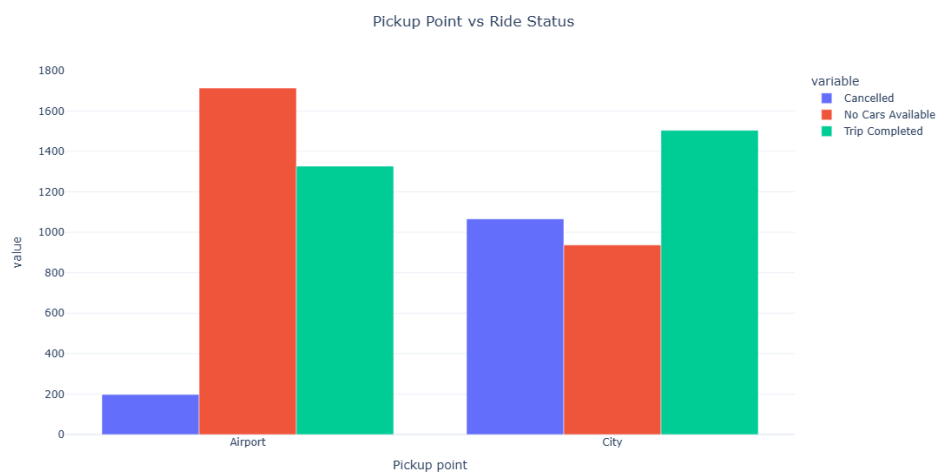
* Cancellation Analysis

Driver cancellations were analyzed to identify periods with higher ride failures.



* Airport vs City Analysis

Ride outcomes were compared across pickup locations.



8. SQL Analysis

SQL was used to validate business findings and generate analytical insights.

Major SQL analyses included:

- Ride Status Analysis
- Demand by Hour Analysis
- Cancellation Analysis
- No Cars Available Analysis
- Pickup Point Comparison

Sample Query

```
SELECT status,  
COUNT(*) AS total_requests  
FROM Muber_requests  
GROUP BY status;
```

The SQL results supported the findings generated through Python analysis.

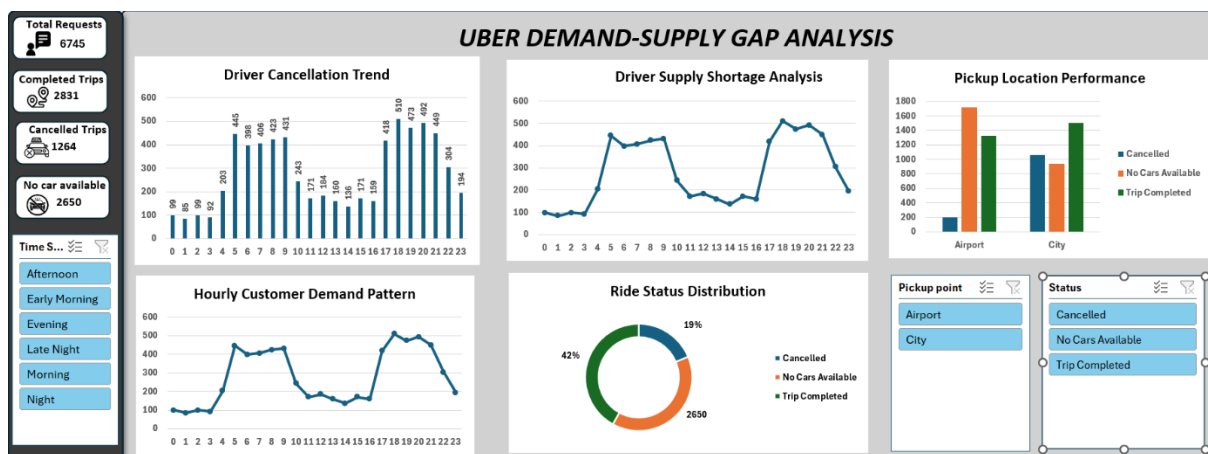
9. Excel Dashboard

An interactive dashboard was developed in Excel using:

- Pivot Tables
- Pivot Charts
- KPI Cards
- Slicers

Dashboard Components:

- Total Requests
- Completed Trips
- Cancelled Trips
- No Cars Available
- Hourly Demand Analysis
- Pickup Point Analysis



The dashboard provides a consolidated business view of operational performance.

10. Key Business Insights

Insight 1

A significant demand-supply gap exists due to insufficient driver availability.

Insight 2

Ride requests increase sharply during peak hours.

Insight 3

Airport pickup locations experience more supply shortages.

Insight 4

Driver cancellations contribute significantly to ride failures.

Insight 5

A large percentage of ride requests do not convert into completed trips.

11. Business Recommendations

Recommendation 1

Increase driver availability during peak demand hours.

Recommendation 2

Improve driver allocation near airport pickup locations.

Recommendation 3

Introduce driver incentive programs to reduce cancellations.

Recommendation 4

Implement smart driver allocation based on demand forecasting.

Recommendation 5

Provide peak-hour bonuses to encourage driver participation.

12. Conclusion

The Uber Demand-Supply Gap Analysis successfully identified the major factors affecting ride completion rates. Driver shortages, particularly during peak hours and at airport locations, were found to be the primary causes of ride failures. Driver cancellations further reduced customer satisfaction and operational efficiency.

The implementation of improved driver allocation strategies, incentive programs, and demand forecasting techniques can significantly reduce the demand-supply gap, improve ride completion rates, and enhance customer experience.

This project demonstrates the practical application of Python, SQL, and Excel in solving real-world business problems through data analytics.